Performance Testing

Date	15 March 2026
Team ID	SWTID-2026-2085
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Apache JMeter
Type of Testing	Load Testing
Target Module / API	Crop Recommendation Prediction (Flask API / Predict Module)
Test Environment	Local System (Windows 11, Python Flask Server)
Test Date	15 March 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Single user submits crop prediction request	1	30	Recommendation displayed successfully
2	Multiple users access the prediction page simultaneously	20	60	Application responds without errors

3	Continuous crop prediction requests	50	120	Stable response with acceptable response time
4	Peak load testing	100	180	Stable response with acceptable response time

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Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds			
2	Response Time (Max)	< 5 seconds			
3	Throughput (Req/sec)				
4	Error Rate	< 1%			
5	CPU Utilization	< 80%			
6	Memory Utilization	< 80%			

Step 4: Observations & Analysis

Key Findings:

[Describe what the test results revealed about your system's performance]

Bottlenecks Identified:

[List any performance bottlenecks observed during testing]

Optimization Steps Taken:

[Describe any changes or optimizations made to improve performance]

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):